

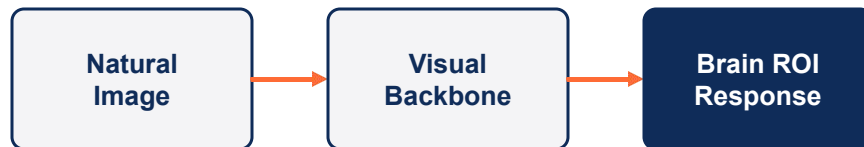
ROI- and Semantics-Aware Feature Pruning for Brain-Aligned Visual Encoding

Computer Vision Project

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Problem Statement

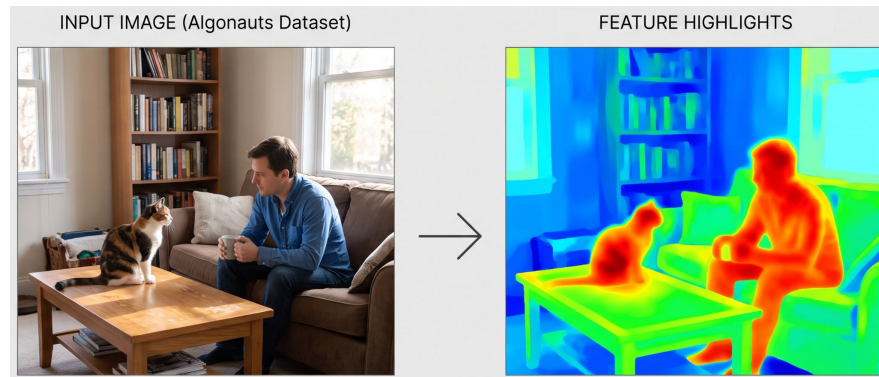
Modern visual backbones produce rich but high-dimensional feature representations.

- **Pretrained Visual Models:** DINOv2 and CLIP provide exceptionally strong pretrained visual representations.
- **Dimensionality Barrier:** These generated representations are extremely high-dimensional, making them difficult to interpret.
- **Functional Specialization:** Different cortical ROIs (Regions of Interest) may rely on distinct subsets of visual information.
- **Feature Pruning Concept:** We ask whether compact ROI-specific feature subsets can fully preserve fMRI brain prediction performance.



Key Question:

Can we reduce visual features while preserving ROI-level brain-encoding performance?



State of the Art and Research Inspiration

This project is inspired by recent work on brain-aligned visual encoders and compact neural representations.

VISGate:

- Uses ROI-conditioned encoders
- The visual representation is adapted to the target brain ROI
- Main idea: different ROIs may require different visual features.

BOLDREAMS:

- Studies pruned in-silico fMRI encoding models.
- Uses pretrained visual backbones to predict visual cortex responses.
- Explores compact “maximally minimal” models and interpretability tools.

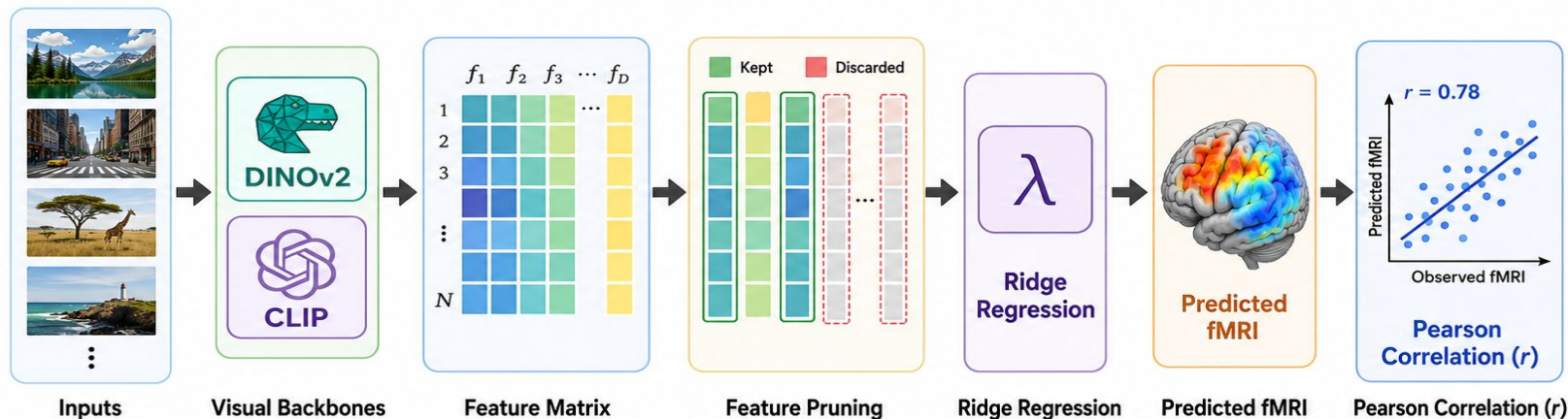
Our Work

- Uses frozen DINOv2 and CLIP features.
- Compares full features, random pruning, global pruning, ROI-aware pruning and semantics-aware pruning.
- Focuses on a simple and reproducible Ridge Regression encoding pipeline.
- Adds semantic grouping to link image categories with ROI response patterns.

Proposed Method: Full Pipeline

The notebook implements a complete visual encoding pipeline from raw inputs to evaluations.

- **Feature Extraction:** Extract frozen high-dimensional visual representations from DINOv2 and CLIP.
- **Pruning Strategies:** Select optimal feature subsets to reduce dimensionality and isolate key functional zones.
- **Ridge Regression:** Train linear encoding models to map features directly to fMRI brain responses.
- **Evaluation:** Benchmark predictions against actual ROI-level fMRI responses using Pearson correlation.



Dataset and fMRI Targets

We model brain-aligned responses using curated visual cortex datasets mapped to semantic ROI structures.

Dataset:

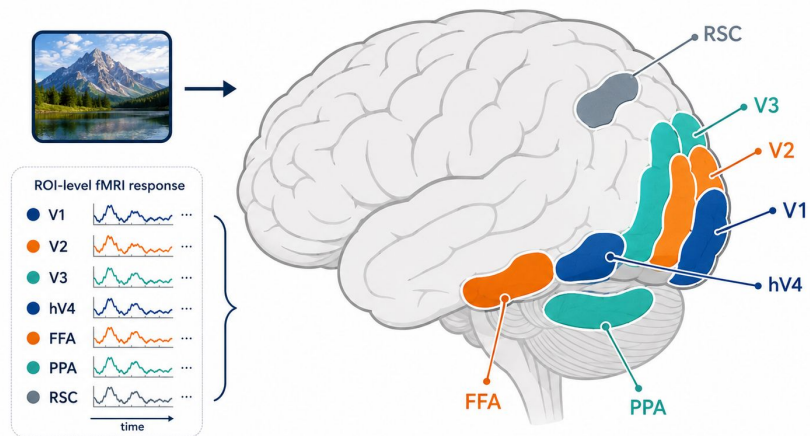
- Natural images from an **NSD / Algonauts** visual cortex dataset.
- Each image is associated with human fMRI responses at ROI level.

COCO Integration:

- NSD images are matched to their original COCO IDs to recover panoptic semantic categories.
- COCO masks are crop-corrected to match the actual stimulus shown to the subject.
- Visible objects and stuff categories are converted into broader semantic groups (people, animals, food, vehicles, furniture, tools, sports objects, ...)

Target Definition

- The model predicts exactly one scalar response per ROI.
- Bypasses unstable voxel prediction for robust, interpretable visual encoding.



Experimental Setup

Structured parameters and evaluation configurations for the brain-encoding pipeline.

Component	Configuration
Backbones	DINOv2 and CLIP, frozen feature extractors
Regressor	Ridge Regression
Training subject	Subject 01
Evaluation 1	Within-subject: train and test on Subject 01
Evaluation 2	Cross-subject: train on Subject 01, test on Subjects 02–05
Keep rations	0.10, 0.25, 0.50, 0.75, 1.00
Main metric	Pearson correlation
Uncertainty metric	Bootstrap Confidence Interval 95%

Feature Pruning Strategies

We compare full feature representations with several feature selection strategies to identify optimal compact subsets.

01

Full Features

Uses all extracted features without any pruning or selection.

Baseline representation

02

Random Pruning

Keeps a completely random subset of features to serve as a strict control baseline.

03

Global Variance

Keeps features with high global variance. Implements a generic pruning baseline.

04

PROPOSED

ROI-Aware

Selects features based on ROI-specific Ridge coefficients. Our main proposed method for extreme compactness.

05

PROPOSED

Semantics-Aware

Uses semantic image groups to guide feature selection and interpret localized ROI responses.

Model Evaluation

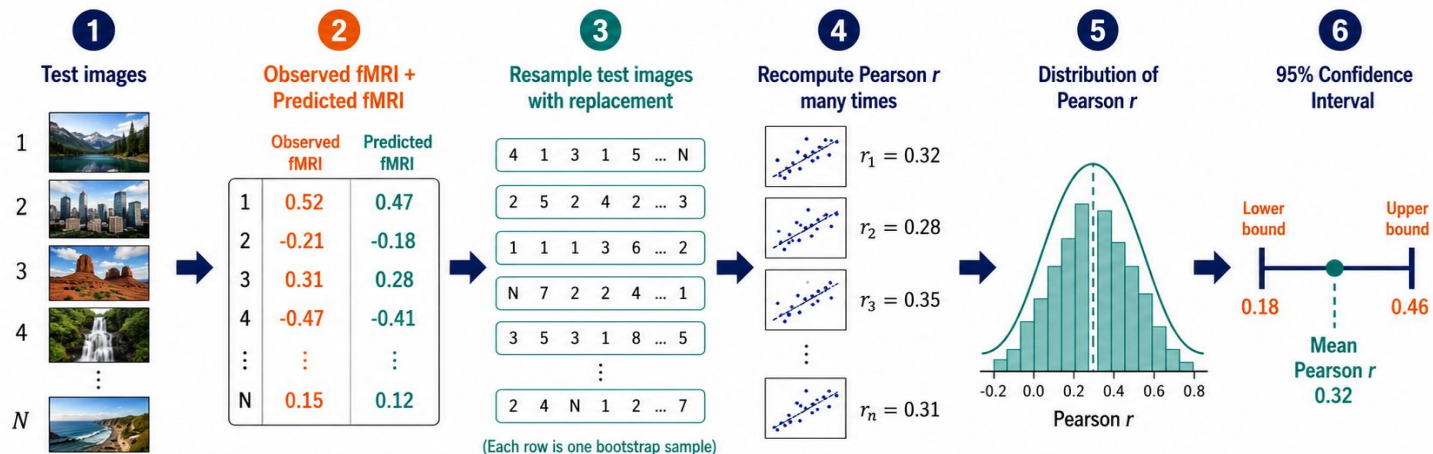
Main metric: Pearson correlation between predicted and true ROI-level fMRI responses.

Higher Pearson r : Indicates superior alignment between model-driven predictions and empirically measured BOLD fMRI responses.

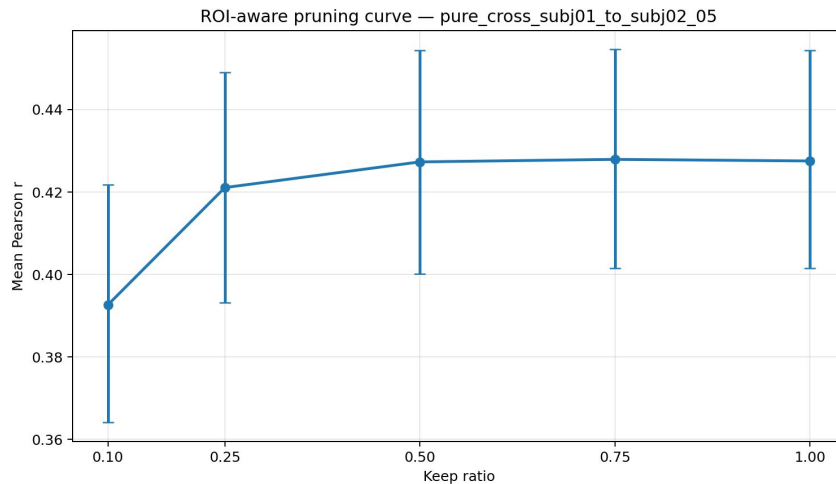
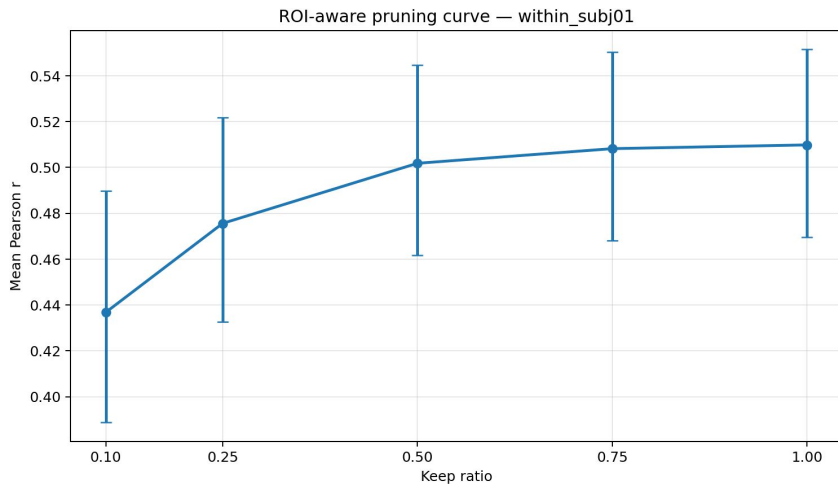
Bootstrap Confidence Intervals: Rigorously estimates prediction uncertainty by non-parametrically resampling the set of test stimuli.

Baseline Comparison: Overall prediction accuracy is systematically benchmarked against the unpruned, full-feature model.

Retained Performance: Quantifies the precise percentage of baseline visual encoding capabilities preserved after aggressive feature pruning.



ROI-aware Pruning Curves



Key observations:

- Performance increases systematically with the feature keep ratio.
- At a **50% keep ratio**, ROI-aware pruning is already extremely close to full-feature baseline performance.
- Cross-subject generalization performance is especially stable and robust at the **25–50% keep ratio** range.

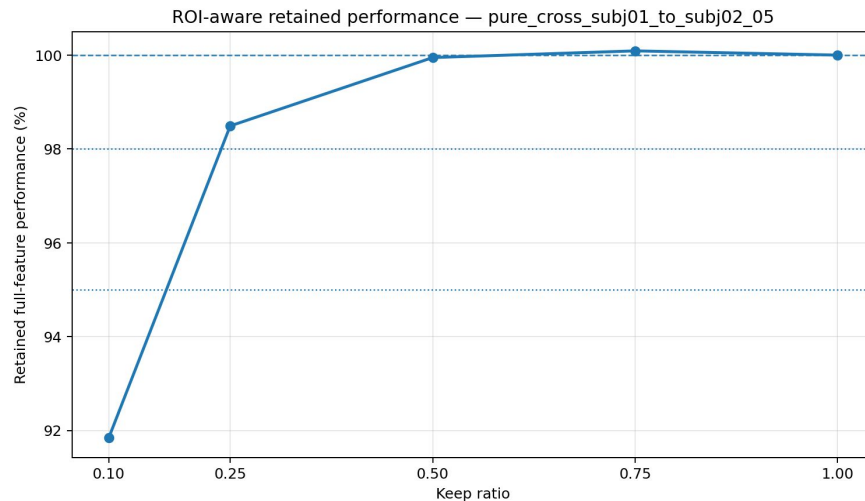
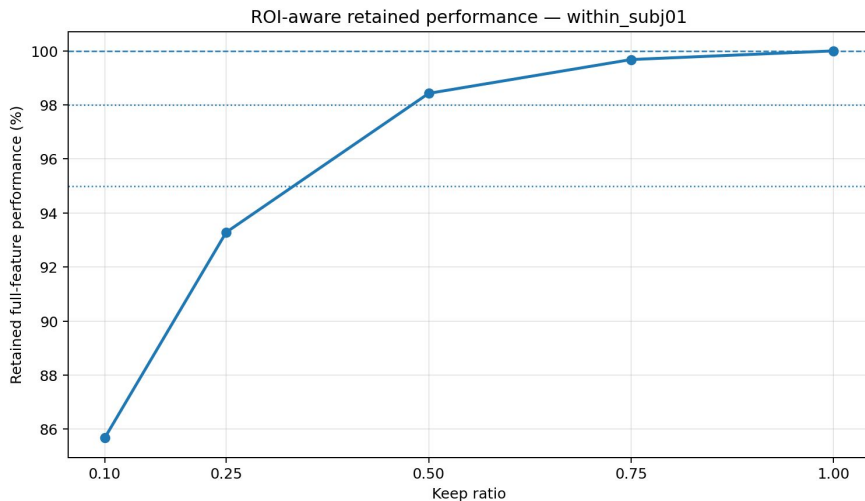


Final Takeaway

"Half of the features are enough to preserve almost the full ROI-level encoding performance."

Retained Performance vs Full Features

Comparing performance retention relative to the unpruned baseline across different keep ratios.



Metric Explanation:

In the cross-subject evaluation, a pruned configurations slightly exceed 100%. This can happen because feature pruning removes redundant or subject-specific noisy dimensions, acting as an additional regularization mechanism. However, the difference is very small, so we should not overclaim it as a clear improvement. The main result remains that pruning preserves almost all full-model performance with fewer features.

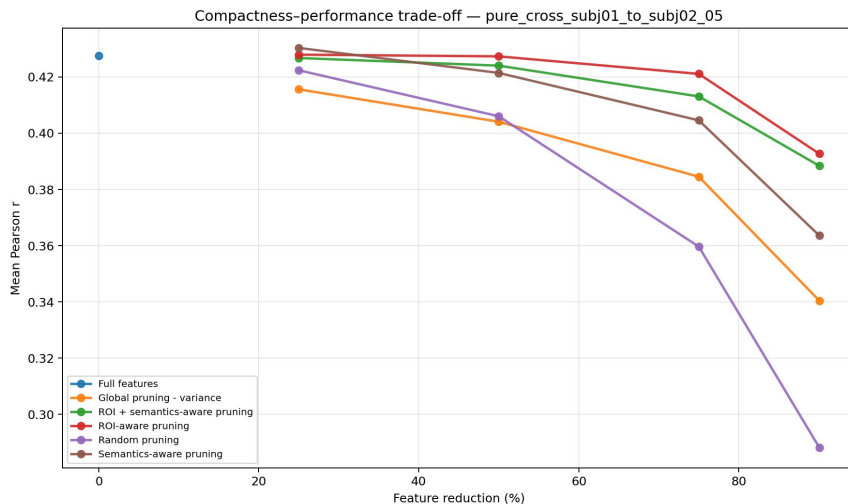
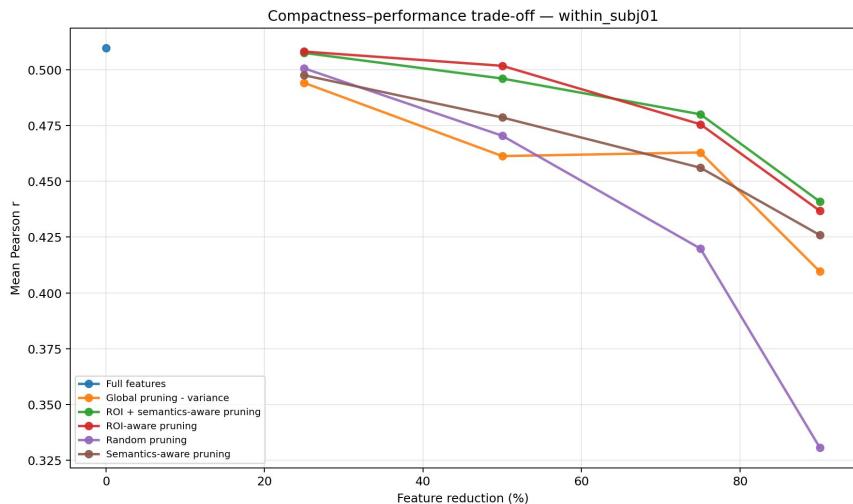
Key Finding



ROI-aware 50% retains 98.43% within-subject and 99.95% cross-subject performance.

Compactness-Performance Trade-off

Analyzing the balance between aggressive feature reduction and visual encoding performance retention.



Interpretation

“The contribution is not a higher absolute Pearson r , but a better compactness-performance trade-off.”

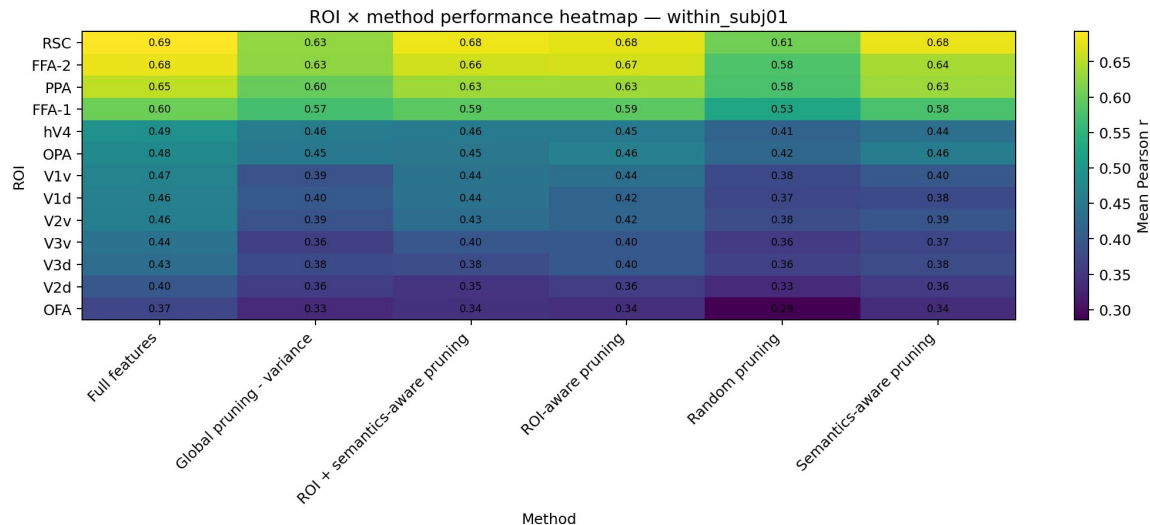
ROI-aware pruning successfully minimizes feature dimensionality while ensuring minimal performance degradation in the target brain regions.

ROI-aware 50% Key Metrics

- **50% feature reduction** achieved successfully.
- **98.43%** retained performance **within-subject**.
- **99.95%** retained performance **cross-subject**.

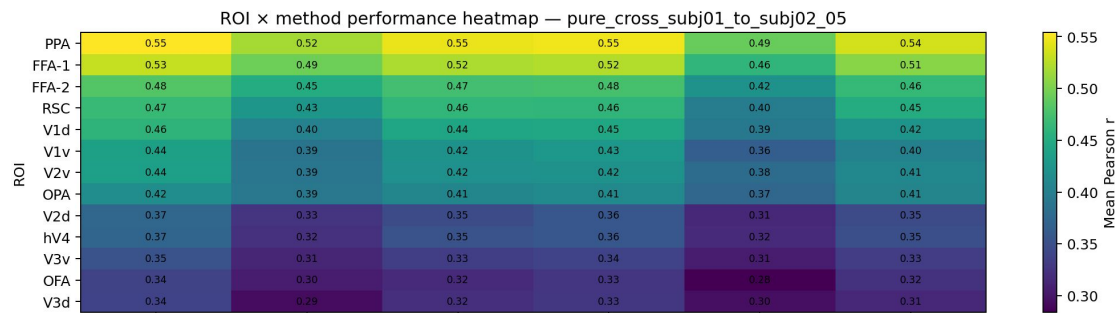
ROI-level Analysis

Detailed examination of feature pruning strategies across individual functional regions of interest.



Interpretation

- Different ROIs show **different prediction difficulty**, indicating varying baselines across visual regions.
- ROI-aware pruning is motivated by the fact that each functional ROI may rely on **distinct visual feature subsets**.
- The performance degradation or preservation effect of pruning is **not uniform** across cortical regions.

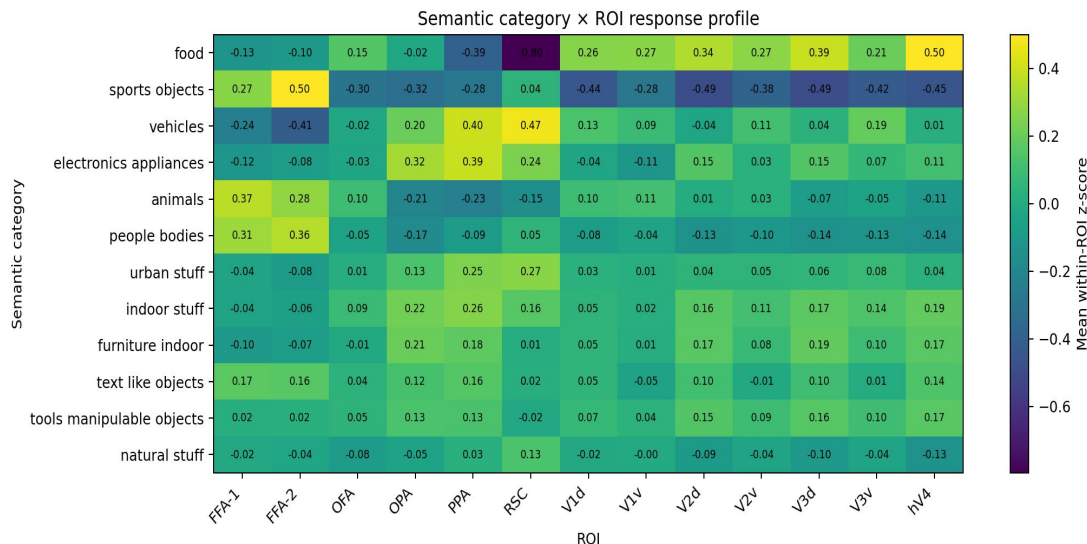


Key Takeaway

“ROI-specific feature selection is justified because visual ROIs do not behave identically.”

Semantic-aware ROI Response Analysis

Exploring localized cortical preferences and visual category processing across key brain regions.



Semantic Groups	Best ROI	Interpretation
animals / people	FFA-1 / 2	high-level visual category preference
vehicles / urban	RSC	scene or navigation-related preference
indoor / furniture	PPA / OPA	place-related preference
food / tools	V3d / hV4	object-related visual response

Conclusions and Future Work

Synthesizing findings on compact, brain-aligned visual encoding and mapping next steps.

Conclusions

1. Pretrained visual features can effectively predict ROI-level fMRI responses.
2. ROI-aware pruning preserves almost all full-feature performance with fewer features.
3. At a 50% keep ratio, ROI-aware pruning retains about 98–100% of full-model performance.
4. Semantics-aware analysis is primarily useful for interpretability.
5. The project successfully supports compact and interpretable brain-aligned visual encoding.

Future Work

- Transition from ROI-level to voxel-level prediction
- Compare additional backbones and layers (DINOv2, CLIP)
- Integrate subject-aware adaptation and estimate noise ceiling
- Incorporate rigorous statistical significance tests



Key Takeaway

ROI-aware pruning does not clearly outperform full features, but it can remove a large fraction of features while preserving almost the same brain-encoding performance.

Key References

- VISGate: ROI-Conditioned Dual-Head Encoders
- BOLDREAMS: Dreaming with pruned in-silico fMRI models
- DINOv2 & CLIP Backbones
- Natural Scenes Dataset / Algonauts-style brain data / COCO Dataset